

Ankith Mohan

Postdoctoral Research Associate

Manning College of Information and Computer Sciences, University of Massachusetts Amherst

Amherst, MA 01003 | ankithmohan@umass.edu | ankith-mohan.github.io

Research Interests

Quantum information and computation; quantum networks and entanglement distribution; reinforcement learning for quantum systems; quantum state discrimination; quantum error correction and optimization.

Employment

Postdoctoral Research Associate

Aug. 2026–present

University of Massachusetts Amherst, College of Information and Computer Sciences

Advisors: Gayane Vardoyan and Philip S. Thomas.

Education

PhD, Computer Science

2026

Virginia Tech, Blacksburg, VA

Advisor: Jamie Sikora.

Dissertation: *Techniques for computing and approximating large optimization problems in quantum information.*

MS, Computer Science

2024

Virginia Tech, Blacksburg, VA

MS, Computer Science

2020

University of Southern California, Los Angeles, CA

Advisors: Aiichiro Nakano and Emilio Ferrara.

Thesis: *Alleviating the Noisy Data Problem using Restricted Boltzmann Machines.*

BE, Information Science and Engineering

2016

Ramaiah Institute of Technology, Bengaluru, India

Research Experience

Graduate Research Assistant

Summer 2025

Virginia Tech, Blacksburg, VA

Advisor: Sumeet Khatri.

Research Intern

Summer 2024

Agency for Science, Technology and Research (A*STAR), Singapore

Advisor: Kishor Bharti.

Research Intern

Summers 2022, 2023

Fujitsu Research of America, Sunnyvale, CA

Advisor: Sarvagya Upadhyay.

Graduate Research Assistant

Spring 2019

USC Information Sciences Institute, Marina del Rey, CA

Advisors: Robert F. Lucas and Jeremy Liu.

Research Assistant

2018–2020

University of Southern California, Los Angeles, CA

Advisor: Sze-Chuan Suen.

Publications

Peer-reviewed articles and proceedings

- [6] **Ankith Mohan**, Tobias Haug, Kishor Bharti, and Jamie Sikora. “Quantum heuristics for linear optimization over large separable operators.” *IEEE TPS-ISA*, pp. 580–589, 2025 [\[link\]](#)
- [5] Mohammad Beigi, Ying Shen, Runing Yang, Zihao Lin, Qifan Wang, **Ankith Mohan**, Jianfeng He, Ming Jin, Chang-Tien Lu, and Lifu Huang. “InternalInspector I^2 : Robust confidence estimation in LLMs through internal states.” *Findings of EMNLP*, pp. 12847–12865, 2024 [\[link\]](#)
- [4] Caleb McIrvine, **Ankith Mohan**, and Jamie Sikora. “Quantum state exclusion through offset measurement.” *Physical Review A* 110, 042211, 2024 [\[link\]](#)
- [3] **Ankith Mohan**, Aiichiro Nakano, and Emilio Ferrara. “Graph signal recovery using restricted Boltzmann machines.” *Expert Systems with Applications* 185, 115635, 2021 [\[link\]](#)
- [2] Jeremy Liu, **Ankith Mohan**, Rajiv K. Kalia, Aiichiro Nakano, Ken-ichi Nomura, Priya Vashishta, and Ke-Thia Yao. “Boltzmann machine modeling of layered MoS₂ synthesis on a quantum annealer.” *Computational Materials Science* 173, 109429, 2020 [\[link\]](#)
- [1] Krishnaraj P. M., **Ankith Mohan**, and Srinivasa K. G. “Performance of procedures for identifying influentials in a social network: Prediction of time and memory usage as a function of network properties.” *Social Network Analysis and Mining* 7, 34, 2017 [\[link\]](#)

Preprints and manuscripts

- [4] Tathagata Gupta, **Ankith Mohan**, Shayeef Murshid, Vincent Russo, Jamie Sikora, and Alice Zheng. “Local strategies are pretty good at computing Boolean properties of quantum sequences.” *arXiv:2603.05452*, 2026 [\[link\]](#)
- [3] Nirupam Basak, **Ankith Mohan**, Andrew Tanggara, Tobias Haug, Goutam Paul, and Kishor Bharti. “Hierarchical quantum decoders.” *arXiv:2601.21715*, 2026 [\[link\]](#)
- [2] Nirupam Basak, Andrew Tanggara, **Ankith Mohan**, Goutam Paul, and Kishor Bharti. “Approximate dynamical quantum error-correcting codes.” *arXiv:2502.09177*, 2025 [\[link\]](#)
- [1] **Ankith Mohan**, Jamie Sikora, and Sarvagya Upadhyay. “A hybrid quantum-classical algorithm for Bayes-optimal quantum state discrimination using the source code.” *arXiv:2312.04023*, updated 2026 [\[link\]](#)

Book

Krishnaraj P. M., **Ankith Mohan**, and Srinivasa K. G. *Practical Social Network Analysis with Python*. Springer.

2018

Patents

- **Ankith Mohan** and Sarvagya Upadhyay. “Hybrid Classical-Quantum Unsupervised Multiclass Classification.” [US20250094447A1](#); granted March 2026. 2026
- Xiaoyuan Liu, **Ankith Mohan**, and Sarvagya Upadhyay. “Identification of Influential Nodes in Graph Datasets Using Combinatorial Optimization Formulations.” [US12437003B2](#); granted October 2025. 2025

Selected Presentations

Talks

- “Hierarchical quantum decoders.” Applied Algebra Seminar, Virginia Tech. Apr. 2026
- “Hybrid Classical-Quantum Unsupervised Multiclass Classification.” Fujitsu Research of America seminar. Jul. 2023
- “NISQ Algorithms for Separable Ground States.” Virginia Tech Quantum Information Science Symposium. Apr. 2022
- “Approximating the Influence Maximization Problem in Social Networks Using DA and QAOA.” Fujitsu Research of America seminar. Jul. 2022

Posters

- “The pretty bad measurement and optimal bounds for antidistinguishability.” Quantum Information Processing (QIP). Feb. 2025
- “Approximate Dynamical Quantum Error-Correcting Codes.” CCI Student Researcher Showcase. Mar. 2024
- “Pretty bad measurement.” CCI Student Researcher Showcase; **People’s Choice Best Poster Award**. Mar. 2024
- “A generalized framework for quantum state discrimination, hybrid algorithms, and the quantum change point problem.” Virginia Tech Quantum Information Science Symposium. Nov. 2023
- “Inner Approximations and a NISQ Algorithm for the Quantum Separability Problem.” Quantum Information Processing (QIP). Mar. 2022

Honors and Awards

- Kafura Doctoral Graduate Fellowship, Department of Computer Science, Virginia Tech. 2026
- People's Choice Best Poster Award, CCI Student Researcher Showcase. 2024
- Best Research Award, Department of Computer Science, University of Southern California. 2021

Teaching

Substitute Lecturer

Introduction to Problem Solving in Computer Science Spring 2024
Virginia Tech
Taught networking to two sections of 70 students each.

Graduate Teaching Assistant

Virginia Tech

- **Data and Algorithms Analysis** Spring 2025, Spring 2026
- **Ethics and Professionalism in Computer Science** Fall 2025
- **Introduction to Problem Solving in Computer Science** Fall 2021, Spring 2024

Outreach

MassImpact

Volunteer: Jones Library, Amherst, MA 2026

Virginia Tech Summer Camps

Lead Instructor 2025
C-Tech² summer camp: led an introductory quantum computing session for 64 students.

Volunteer 2025

- Explore Science (Grades 7–8)
- Explore Science (Grades 9–10)
- Explore Life Science
- WEE VT
- Virginia Tech QISE Summer School
- Explore Physical Sciences

Academic Service

Reviewer

- Journal of Physics A: Mathematical and Theoretical
- Quantum Science and Technology
- npj Quantum Information

(Sub-)reviewer

Theory of Quantum Computation, Communication and Cryptography (TQC).

2025

References

Gayane Vardoyan

University of Massachusetts Amherst
Postdoctoral advisor
gvardoyan@cs.umass.edu

Jamie Sikora

Virginia Tech
Doctoral advisor
sikora@vt.edu

Sarvagya Upadhyay

Fujitsu Research of America
Research internship advisor
supadhyay@fujitsu.com

Philip S. Thomas

University of Massachusetts Amherst
Postdoctoral advisor
pthomas@umass.edu

Sumeet Khatri

Virginia Tech
Doctoral committee member
skhatri@vt.edu